

SODS Final Project - Group 17

MURTHI GERALD s343454 ARAPI GERSI s337081 ARAMNIA ALI s338956

1 Baseline Synthesis

The baseline synthesis was performed using the STcmos65 LVT library. Different clock periods, with a step of 0.5 ns, were tested. The minimum clock period satisfying both constraints, slack greater than or equal to zero and area lower than $33300 \mu m^2$, was found to be 2.5 ns.

Table 1: Baseline synthesis results

Metric	Value
Clock period (ns)	2.500000
Area (μm^2)	33207.200083
Slack (ns)	0.000099
Leakage power (mW)	0.01244
Dynamic power (mW)	15.846

The power values were measured in PrimeTime using VCD-based switching activity back-annotation. The VCD annotation reported 100% annotated nets and 100% fully annotated leaf cells.

2 Power Optimization

2.1 Clock-Gating

Clock gating was applied starting from the clock period found in the baseline synthesis, equal to 2.5 ns. The clock-gating style was configured with minimum bit-width equal to 8 and maximum fanout equal to 32.

Table 2: Clock-gating optimization results

Metric	Value
Area (μm^2)	30000.880089
Slack (ns)	0.000028
Leakage power (mW)	0.01212
Dynamic power (mW)	19.597

Discussion. After applying clock gating, the area decreased from $33207.200083 \mu m^2$ to $30000.880089 \mu m^2$. The design still satisfies the timing constraint, with a positive slack of 0.000028 ns. The leakage power slightly decreased from 0.01244 mW to 0.01212 mW, mainly because the final synthesized design has a smaller cell area.

The dynamic power increased from 15.846 mW to 19.597 mW in this implementation. Although clock gating is generally expected to reduce dynamic power by disabling unnecessary clock activity, the final result also depends on the new synthesized structure, the inserted clock-gating logic, and the switching activity captured in the VCD file. In this run, the reduction of area and leakage was obtained, but the total dynamic power did not improve.

2.2 Clock-Gating + Multi-VT

Clock gating was combined with Multi-VT synthesis using LVT, SVT, and HVT libraries. The same clock period of 2.5 ns was used. The same clock-gating settings of Exercise 2.1 were kept.

Table 3: Clock-gating and Multi-VT optimization results

Metric	Value
Percentage of LVT cells (%)	38.508867
Percentage of SVT cells (%)	18.747738
Percentage of HVT cells (%)	42.743395
Area (μm^2)	37878.360001
Slack (ns)	0.000192
Leakage power (mW)	0.006853
Dynamic power (mW)	20.131

Discussion. The Multi-VT synthesis selected a mixed implementation composed of 38.508867% LVT cells, 18.747738% SVT cells, and 42.743395% HVT cells. The large percentage of HVT cells reduced the leakage power from 0.01212 mW in the clock-gated solution to 0.006853 mW. This is the expected effect of HVT cells, since they have lower leakage current.

The timing constraint is still satisfied, with a positive slack of 0.000192 ns. However, the area increased to 37878.360001 μm^2 . This can happen because the synthesis tool uses different threshold-voltage versions and additional cells to satisfy timing while replacing part of the design with lower-leakage cells. The dynamic power also increased slightly to 20.131 mW. Therefore, in this solution the main benefit of Multi-VT synthesis is the leakage-power reduction, while the cost is an increase in area and dynamic power.